POLITECNICO
MILANO 1863

# The Harrow-Hassidim-Lloyd algorithm

PhD course "The ABCs of Quantum Computing"

**Marco Galliani**

May 27th, 2026

# Context: what is the HHL algorithm



A quantum algorithm for linear systems $Ax = b$, $A \in \mathbb{R}^{N \times N}$

- HHL does not compute a solution $x$
- "expectation value of some operator associated with $x$, e.g., $x^\top Mx$"
- Logarithmic scaling in the matrix dimension, $\mathcal{O}(\log N\kappa^2/\epsilon)$

# General idea

Let $Ax = b$ be a linear system with $A$ $N_b \times N_b$ matrix and $x$ and $b$ $N_b$-dimensional vectors.
Matrix $A$ can be written a linear combination of outer products of its eigenvectors weighted by its eigenvalues:

$$A = \sum_i \lambda_i \mathbf{u}_i \mathbf{u}_i^\top$$

The inverse matrix $A^{-1}$ can be expressed as

$$A^{-1} = \sum_i \lambda_i^{-1} \mathbf{u}_i \mathbf{u}_i^\top$$

# HHL: steps



1. State preparation
2. Quantum Phase Estimation (QPE)
3. Ancilla qubit rotation
4. Inverse QPE

# Registers I



Little endian convention (righmost qubit is the less significant). Then, from the most significant to the leas significant

1. *b-register*: components of $\mathbf{x}, \mathbf{b}$, encoded as amplitudes/coefficients of basis states of the $n_b$-qubits, with $N_b = 2^{n_b}$
2. *c-register*: values of the phase of the eigenvalues of $A$ after QPE
3. *ancilla*: used for inversion and measurement, discarded in the solution

Initial state:

$$|\Psi_0\rangle = |0\rangle^{\otimes n_b} |0\rangle^{\otimes n} |0\rangle$$

# Registers II



ancilla qubit

c-registers

b-registers

# State preparation I

The state preparation step consists of rotating the qubits in the b-register to have amplitudes corresponding to the coefficients of **b** (*amplitude encoding*)

$$|b\rangle = \beta_0 |0\rangle + \beta_1 |1\rangle + \cdots + \beta_{N_n-1} |N_b - 1\rangle$$

Leading to the updated state:

$$|\Psi_1\rangle = |b\rangle_b |0\rangle^{\otimes n} |0\rangle_a$$

# State preparation II

The amplitude encoding is implemented in the following way:

1. $\mathbf{b}$ is normalized: $\tilde{\mathbf{b}}$ (amplitudes need to sum to 1 in quantum states)
2. $|\tilde{b}\rangle = \tilde{b}_0 |0\rangle + \tilde{b}_1 |1\rangle_1 + \cdots + \tilde{b}_{N_b-1} |N_b - 1\rangle$

Consider the simple example where $\mathbf{b} = (0, 1)$. The vector is already normalized and the state preparation circuit corresponds to the following:

**Note:** The state preparation step is then specific to each $\mathbf{b}$.

The goal of QPE is to estimate the phase of the eigenvalues of $A$
Overview:

1. Superposition
2. Controlled rotations
3. Inverse Quantum Fourier Transforms (IQFT)

The goal of this step is to put the clock qubits into an equal superposition annd explore all possible phases simultaneously

$$
\begin{aligned}
|\Psi_2\rangle &= I^{\otimes n_b} \otimes H^{\otimes n} \otimes I \,|\Psi_1\rangle \\
&= |b\rangle \,\frac{1}{2^{\frac{n}{2}}}(|0\rangle + |1\rangle)^{\otimes n} \,|0\rangle
\end{aligned}
$$

In order to estimate the eigenvalues of $A$, we need it to somehow operate with it within the quantum circuit. As $A$ in non-unitary and gates of a quantum circuit need to be unitary we have to encode $A$ through Hamiltonian encoding, namely we're gonna use the unitary gate

$$U = e^{iAt}$$

**Interpretations:**

- eigenvalue connection: $\mathrm{eig}(U) = e^{i\lambda_i t}$, where $\lambda_i$ are the eigenvalues of $A$
- $A$ energy of the Hamiltonian system

Eigenvalues of $A$ will be stored in the *c-registers*, encoded using *basis encoding*. For this reason, the stored eigenvalues are not exact. In this

section, controlled $U^t$ gates will be applied to $|b\rangle$.

**Motivation:** Assume $|b\rangle$ to be an eigenvector of $U$ with eigenvalues $e^{2\pi i\phi}$. Then

$$U|b\rangle = e^{2\pi i\phi}|b\rangle .$$

When the control quibit is 0 $|b\rangle$ is not affected, otherwise $U$ is applied to it. This relationship is expressed by

$$\begin{aligned}|\Psi_3\rangle &= |b\rangle \otimes (\frac{1}{2^{\frac{n}{2}}}\bigotimes_{j=1}^{n}(|0\rangle + e^{2\pi i\psi 2^{n-j}}|1\rangle)) \otimes |0\rangle_a \\ &= |b\rangle \otimes (\frac{1}{2^{\frac{n}{2}}}\sum_{k=1}^{2^n-1} e^{2\pi i\psi k}|k\rangle) \otimes |0\rangle_a\end{aligned}$$

$$
\begin{aligned}
|\Psi_4\rangle &= |b\rangle \otimes \mathrm{IQFT}\Big(\frac{1}{2^{\frac{n}{2}}}\sum_{k=0}^{2^{n-1}}(e^{2\pi i\psi k}\,|k\rangle)\Big) \otimes |0\rangle_a \\
&= |b\rangle \otimes \Big(\frac{1}{2^{\frac{n}{2}}}\sum_{k=0}^{2^{n-1}}(e^{2\pi i\psi k}\mathrm{IQFT}(|k\rangle))\Big) \otimes |0\rangle_a \\
&= |b\rangle \otimes \Big(\frac{1}{2^{\frac{n}{2}}}\sum_{k=0}^{2^{n-1}}(e^{2\pi i\psi k}(\frac{1}{\sqrt{N}}\sum_{j=0}^{N-1} e^{-2\pi ijk/N}\,|j\rangle))\Big) \otimes |0\rangle_a \\
&= |b\rangle \otimes \Big(\frac{1}{2^{\frac{n}{2}}}\sum_{k=0}^{2^{n-1}}(e^{2\pi i\psi k}(\frac{1}{\sqrt{2^n}}\sum_{j=0}^{N-1} e^{-2\pi ijk/N}\,|j\rangle))\Big) \otimes |0\rangle_a \\
&= |b\rangle \otimes \Big(\frac{1}{2^n}\sum_{k=0}^{2^{n-1}}(\sum_{j=0}^{2^n-1} e^{2\pi ik(\phi-j/N)}\,|j\rangle)\Big) \otimes |0\rangle_a
\end{aligned}
$$

Thus,

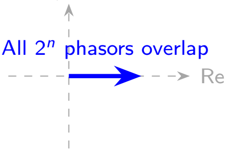
$$|\Psi_4\rangle = |b\rangle \otimes \left(\frac{1}{2^n}\sum_{k=0}^{2^{n-1}}\left(\sum_{j=0}^{2^n-1} e^{2\pi i k(\phi - j/N)}\,|j\rangle\right)\right) \otimes |0\rangle_a$$

and the following altenatives apply:

- if $\phi - j/N = 0$: $|j\rangle$ finite amplitude equal to $\sum_{k=0}^{2^n-1} e^0 = 2^n$, *constructive interference*
- otherwise: $\sum_{k=0}^{2^n-1} e^{2\pi i k(\phi - j/N)} = 0$, *destructive interference*

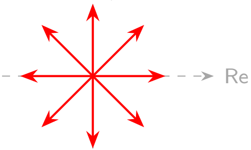
**Constructive Interference**

$\phi - j/N = 0$




**Destructive Interference**

$\phi - j/N \neq 0$



Finally, ignoring the states with zero-amplitude:

$$
\begin{aligned}
|\Psi_4\rangle &= \frac{1}{2^n} |b\rangle \sum_{k=0}^{2^n-1} e^{2\pi i k \cdot 0} |N\phi\rangle |0\rangle_a \\
&= |b\rangle |N\phi\rangle |0\rangle_a
\end{aligned}
$$

In this way, the clock qubits are used to represent the phase information of $U$, the accuracy depends on the number of qubits $n$

# QPE: Inverse Quantum Fourier Tranform IV



Since $U$ encodes $A$ through Hamiltonian encoding $U = e^{iAt}$, $U$ is diagonal in the eigenvectors of $A$:

$$U |b\rangle = e^{i\lambda_j t} |u_j\rangle$$

Equating the latter to $U |b\rangle = e^{2\pi i\phi} |b\rangle$, we get $\phi = \frac{\lambda_j t}{2\pi}$ and

$$|\Psi_4\rangle = |u_j\rangle \, |\frac{N\lambda_j t}{2\pi}\rangle \, |0\rangle_a$$

More generally, expressing $|b\rangle$ in the basis formed by the eigenvectors of $A$, we have

$$\sum_{j=0}^{2_b^n-1} b_j |u_j\rangle \, |\frac{N\lambda_j t}{2\pi}\rangle \, |0\rangle_a$$

As $\lambda_j$ are usually non-integers, $t$ will be selected so that $\frac{N\lambda_j t}{2\pi}$ are integers

Rotation of the ancilla qubit is performed on the basis of a parameter $C$

$$|\Psi_5\rangle = \sum_{j=0}^{2^{n_b}-1} b_j |u_j\rangle |\tilde{\lambda}_j\rangle \left(\sqrt{1 - \frac{C^2}{\tilde{\lambda}_j^2}} |0\rangle_a + \frac{C}{\tilde{\lambda}_j} |1\rangle_a\right)$$

Once the ancilla qubit is measured we got two alternatives

- if $|0\rangle$: repeat the computation
- if $|1\rangle$: stop the iterations

After such iterations, the state is updated to

$$|\Psi_6\rangle = \frac{1}{\sum_{j=0}^{2^{n_b}-1} |\frac{b_j N}{\tilde{\lambda}_j}|^2} \sum_{j=0}^{2^{n_b}-1} b_j |u_j\rangle |\tilde{\lambda}_j\rangle \frac{C}{\tilde{\lambda}_j} |1\rangle_a$$

# Inverse QPE I



$$|\Psi_6\rangle = \frac{1}{\sum_{j=0}^{2^{n_b}-1} |\frac{b_j N}{\tilde{\lambda}_j}|^2} \sum_{j=0}^{2^{n_b}-1} C \frac{b_j}{\tilde{\lambda}_j} |u_j\rangle |\tilde{\lambda}_j\rangle |1\rangle_a$$

The *b-register* now stores the result in the desired form with respect to the eigenvector-basis. What we have to do now is to express the *b-register* through amplitude encoding. Problem is that the latter is entagled with the other qubits making unfeasible to operate such a conversion.

The inverse QPE step allows then to disentagle the states allowing to express the *b-register* through amplitude encoding.

To do so, QFT is applied to the *c-register* qubits:

$$
\begin{aligned}
|\Psi_7\rangle &= \frac{1}{\sum_{j=0}^{2^{n_b}-1} |\frac{b_j N}{\tilde{\lambda}_j}|^2} \sum_{j=0}^{2^{n_b}-1} C \frac{b_j}{\tilde{\lambda}_j} |u_j\rangle \, \mathrm{QFT}(|\tilde{\lambda}_j\rangle) \, |1\rangle_a \\
&= \frac{1}{\sum_{j=0}^{2^{n_b}-1} |\frac{b_j N}{\tilde{\lambda}_j}|^2} \sum_{j=0}^{2^{n_b}-1} C \frac{b_j}{\tilde{\lambda}_j} |u_j\rangle \, \Big(\frac{1}{2^{\frac{n}{2}}} \sum_{y=0}^{2^n-1} e^{2\pi i y \tilde{\lambda}_j / N} \, |y\rangle\Big) \, |1\rangle_a
\end{aligned}
$$

Then controlled rotations with $U^{-1}$ are applied as before

$$
\begin{aligned}
|\Psi_8\rangle &= \frac{1}{\sum_{j=0}^{2^{n_b}-1} |\frac{b_j N}{\tilde{\lambda}_j}|^2} \sum_{j=0}^{2^{n_b}-1} C \frac{b_j}{\tilde{\lambda}_j} |u_j\rangle \left(\frac{1}{2^{\frac{n}{2}}} \sum_{y=0}^{2^n-1} e^{-i\lambda_j t y} e^{2\pi i y \tilde{\lambda}_j / N} |y\rangle\right) |1\rangle_a \\
&= \frac{1}{2^{\frac{n}{2}} \sum_{j=0}^{2^{n_b}-1} |\frac{b_j N}{\tilde{\lambda}_j}|^2} \sum_{j=0}^{2^{n_b}-1} C \frac{b_j}{\tilde{\lambda}_j} |u_j\rangle \sum_{y=0}^{2^n-1} |y\rangle |1\rangle_a \\
&= \frac{C}{2^{\frac{n}{2}} \sum_{j=0}^{2^{n_b}-1} |\frac{b_j N}{\tilde{\lambda}_j}|^2} |x\rangle \sum_{y=0}^{2^n-1} \frac{1}{2^{\frac{n}{2}}} |y\rangle |1\rangle_a
\end{aligned}
$$

The clock-qubits and *b-registers* are now unentagled. By applying the Hadamard gates n the clock qubits we have

$$|\Psi_9\rangle = \frac{C}{\sum_{j=0}^{2^{n_b}-1} |\frac{b_j N}{\tilde{\lambda}_j}|^2} |x\rangle_b |0\rangle_c^{\otimes n} |1\rangle_a$$

where the solution $|x\rangle$ is stored in the *b-registers* successfully.